MedR-Bench Oracle 诊断：数据结构与评估 IO

1. 文件关系

```
diagnosis_957_*.json          GT，键为 PMC ID
oracle_diagnosis.json         模型输出：{ "PMCxxxxx": { "模型名": results对象 } }

评估合并：
  case_data = GT[case_id].copy()
  case_data["id"] = case_id
  case_data["results"] = oracle[case_id][model_name]

输出：
  acc_results/.../PMCxxxxx.json
  reasoning_results/.../PMCxxxxx.json
```

---

2. GT 单病例（`diagnosis_957_*.json` 中每个 PMC 值）

```
{
  "raw_case": "原始病例全文",
  "generate_case": {
    "case_summary": "结构化病例描述",
    "differential_diagnosis": "GT 鉴别/推理讨论",
    "final_diagnosis": "GT 最终诊断说明（段落）",
    "diagnosis_results": "GT 标准答案（短句）"
  },
  "body_category": ["..."],
  "disorder_category": ["..."],
  "receive_time": "...",
  "checked_rare_disease": []
}
```

字段	推理	评估
`generate_case.case_summary`	模型输入	Efficiency、Factuality
`generate_case.final_diagnosis`	—	Efficiency 的推理目标
`generate_case.differential_diagnosis`	—	Completeness GT 拆分（前半）
`generate_case.final_diagnosis`	—	Completeness GT 拆分（后半）
`generate_case.diagnosis_results`	—	Accuracy
`raw_case`	—	不用

---

3. 模型输出 `results`（`oracle_diagnosis.json`）

模型	字段	评估读取
gemini2-ft / qwq	`content`	☑ 推理步 + 答案
deepseek-r1	`thinking_process`	✖（仅 `--model deepseek-r1-thinkingprocess` 时读）
deepseek-r1	`content`	☑ 默认读此
本地 gemini/qwq	`input`, `content`	仅 `content`
本地 deepseek	`input`, `out_reasoning`, `out_answer`	需转成论文格式后再评

`content` 格式：

```
### Resoning:
<step 1> ...
<step 2> ...
### Answer:
最终诊断
```

deepseek 论文格式：

```
{
  "thinking_process": "原始 CoT 自由文本",
  "content": "### Resoning:\n<step 1> ...\n### Answer:\n..."
}
```

推理输入：`generate_case.case_summary` → prompt 模板。

`split_reasoning(text)`（从 `content` 或 `thinking_process` 切预测步）：

1. 去 ` ``` `、取 `### Resoning:` 等标记后内容
2. 截断 `### Answer:` 之前
3. 优先 `<step n>`；否则按空行/换行切
4. 最多 10 步

用途	读取
Eff / Fact / Comp 预测步	`content`（默认）或 `thinking_process`
Accuracy 预测答案	始终 `content` 中 `### Answer` 后

---

4. 评估结果 JSON

评估结果 = **GT 原对象完整 copy**（`raw_case` + `generate_case` + metadata） + `id` + `results` + 指标。

4.1 Reasoning 结果

```
{
  "raw_case": "...",
  "generate_case": {
    "case_summary": "...",
    "differential_diagnosis": "...",
    "final_diagnosis": "...",
    "diagnosis_results": "..."
  },
  "body_category": ["..."],
  "disorder_category": ["..."],
  "receive_time": "...",
  "checked_rare_disease": [],
  "id": "PMC11625232",
  "results": {
    "thinking_process": "...",
    "content": "..."
  },
  "reasoning_eval": [
    {
      "step": "预测步原文",
      "efficiency": "Reasoning | Citation | Repetition | Redundancy",
      "factulity": true,
      "judgment_path": [{ "judgment": "Correct", "keywords_to_search": "None" }]
    }
  ],
  "gt_reasoning_eval": [
    { "step": "Step 1: ...", "hit": true }
  ],
  "efficiency": 0.85,
  "factulity": 0.92,
  "recall": 0.375
}
```

追加字段	含义
`reasoning_eval[]`	每预测步 Eff + Fact
`gt_reasoning_eval[]`	每 GT 步 + 是否被覆盖
`efficiency`	Reasoning 步占比
`factulity`	Reasoning 步中事实正确占比
`recall`	Completeness

非 Reasoning 步：`factulity: null`, `judgment_path: []`。

4.2 Accuracy 结果

同上 GT 结构 + `results` + 顶层 `accuracy`，**无** reasoning 相关字段。

```
{
  "...GT 字段同上...",
  "id": "PMC11625232",
  "results": { "content": "..." },
  "accuracy": true
}
```

---

5. 各评估步骤 IO

指标	输入	输出
Accuracy	`generate_case.diagnosis_results` `results.content` → Answer 段	`accuracy`
Efficiency	每预测步 + `generate_case.case_summary` + `generate_case.final_diagnosis`	`reasoning_eval[].efficiency` `efficiency`
Factuality	仅 Reasoning 步 + `generate_case.case_summary` + 搜索摘要	`reasoning_eval[].factulity` `factulity`
Completeness	GT：`differential_diagnosis + "\n Final diagnosis:\n" + final_diagnosis` → Judge 切步 Pred：`"\n".join(预测步)`	`gt_reasoning_eval[]` `recall`

Prompt 占位符

Accuracy（`acc_diagnose.txt`）

占位符	来源
`{pred_diagnose}`	`results.content` Answer 段
`{gt_diagnose}`	`generate_case.diagnosis_results`

Efficiency（每预测步 1 次）

占位符	来源
`{current_step}`	当前预测步
`{previous_steps}`	之前各步拼接
`{case}`	`generate_case.case_summary`
`{result}`	`generate_case.final_diagnosis`

Factuality（仅 Reasoning 步）

占位符	来源
`{case}`	`generate_case.case_summary`
`{reasoning_step}`	当前预测步
`{info}`	Bing 摘要或关搜索时的固定占位文本

Completeness — GT 拆分（1 次/例）

占位符	来源
`{gt_reasoning}`	`differential_diagnosis + "\n Final diagnosis:\n" + final_diagnosis`

Completeness — hit-check（每 GT 步 1 次）

占位符	来源
`{a_reasoning_step}`	GT 单步
`{out_reasoning}`	预测步 join 后的字符串

---

6. 字段 → 指标

```
generate_case.case_summary       → 推理输入；Eff/Fact {case}
generate_case.final_diagnosis    → Eff {result}
generate_case.differential_diagnosis ┐
generate_case.final_diagnosis        ├→ Completeness GT 步
results.content                  → split → Eff/Fact/Comp 预测链
results.thinking_process         → 仅 deepseek-r1-thinkingprocess 模式
generate_case.diagnosis_results ← results.content Answer → Accuracy
```

---

7. 完整示例：PMC11625232（Reasoning: deepseek-r1 · Accuracy: gemini2-ft）

以真实数据走一遍：**GT → 推理 → 合并 → 四项评估 → 落盘**。

Step 1 · GT（`diagnosis_957_*.json["PMC11625232"]`）

```
{
  "raw_case": "Neurological and orbital complication of acute sinusitis...（原文，评估不用）",
  "generate_case": {
    "case_summary": "- **Patient Information:** 13-year-old male\n- **Chief Complaint:** Severe left eye pain\n- **History of Present Illness:** Eyelid edema, erythema... recent nasal congestion and headaches.\n- **Physical Examination:** Febrile (39.5°C)...\n- **Ancillary Tests:** Elevated CRP 306.9 mg/L; NCCT/NCMRI: maxillary sinusitis, orbital cellulitis, frontal epidural empyema.",
```

```
    "differential_diagnosis": "1. **Orbital Cellulitis:** ... 2. **Epidural
Empyema:** ... 3. **Acute Sinusitis:** ... 4. **Other Conditions:** excluded ...",
    "final_diagnosis": "The patient was diagnosed with **orbital cellulitis
secondary to acute sinusitis**, complicated by a small **frontal epidural
empyema**.",
    "diagnosis_results": "Orbital cellulitis secondary to acute sinusitis with
frontal epidural empyema."
  },
  "body_category": ["Brain and Nerves"],
  "disorder_category": ["Infections"],
  "receive_time": "2024-10-9",
  "checked_rare_disease": []
}
```

Step 2 · 推理（模型只看 `case_summary`）

```
输入  → generate_case.case_summary  （填入诊断 prompt）
输出  → oracle_diagnosis.json["PMC11625232"]["deepseek-r1"]
```

```
{
  "thinking_process": "Okay, let's start by looking at the patient's information.
He's a 13-year-old male with severe left eye pain... Putting it all together:
orbital cellulitis secondary to acute bacterial sinusitis, complicated by frontal
epidural empyema.",
  "content": "### Resoning:\n<step 1> The patient presents with severe left eye
pain, eyelid edema, erythema... severe bacterial infection.\n<step 2> ... imaging-
confirmed left maxillary sinusitis ... sinus origin.\n<step 3> NCCT and NCMRI
reveal ... frontal epidural empyema.\n<step 4> The progression from sinusitis to
orbital cellulitis ... bacterial sinusitis.\n<step 5> ... ruling out isolated
preseptal cellulitis or non-infectious causes.\n\n### Answer:\nOrbital cellulitis
secondary to acute bacterial sinusitis complicated by frontal epidural empyema."
}
```

`split_reasoning(content)` → **5 个预测步**（评估读 `content`，不读 `thinking_process`）。

Step 3 · 评估前合并

```
case_data = GT["PMC11625232"].copy()
case_data["id"] = "PMC11625232"
case_data["results"] = oracle["PMC11625232"]["deepseek-r1"]
```

Step 4 · 各指标读什么

指标	本例实际取值

指标	本例实际取值
Accuracy	GT: `"Orbital cellulitis secondary to acute sinusitis with frontal epidural empyema."` Pred（Step 6 · gemini2-ft 真跑）：`"Orbital Cellulitis with Frontal Epidural Empyema secondary to Maxillary Sinusitis."`
Efficiency ×5	每步 + `case_summary` + `final_diagnosis`（段落）
Factuality	5 步均判 Reasoning → 逐步评事实性
Completeness	GT 拼接: `differential_diagnosis + "\n Final diagnosis:\n" + final_diagnosis` Pred 链: 5 步 join 成一段字符串

Completeness GT 拆分 → **8 条 GT 步**，逐条与预测链比 hit。

Step 5 · Reasoning 评估落盘

`reasoning_results_qwen_judge_paper_957/deepseek-r1/PMC11625232.json`（截断）：

评估结果是 **GT 整条原样保留**（含 `raw_case`、metadata）+ 追加字段：

```
{
  "raw_case": "Neurological and orbital complication of acute sinusitis...（原文全文，评估不读）",
  "generate_case": {
    "case_summary": "- **Patient Information:** 13-year-old male ...",
    "differential_diagnosis": "1. **Orbital Cellulitis:** ...",
    "final_diagnosis": "The patient was diagnosed with **orbital cellulitis secondary to acute sinusitis** ...",
    "diagnosis_results": "Orbital cellulitis secondary to acute sinusitis with frontal epidural empyema."
  },
  "body_category": ["Brain and Nerves"],
  "disorder_category": ["Infections"],
  "receive_time": "2024-10-9",
  "checked_rare_disease": [],
  "id": "PMC11625232",
  "results": { "thinking_process": "...", "content": "### Resoning:\n<step 1> ..." },
  "reasoning_eval": [
    { "step": "The patient presents with severe left eye pain...", "efficiency": "Reasoning", "factulity": true, "judgment_path": [{"judgment": "Correct", "keywords_to_search": "None"}] },
    { "step": "The recent severe nasal congestion...", "efficiency": "Reasoning", "factulity": true, "judgment_path": [...] },
    "... 共 5 步，均为 Reasoning + factulity=true ..."
  ],
  "gt_reasoning_eval": [
    { "step": "Step 1: Orbital Cellulitis was strongly suspected ...", "hit": true },
```

```
    { "step": "Step 2: Acute Maxillary Sinusitis ...", "hit": false },
    { "step": "Step 3: Small Epidural Empyema ...", "hit": false },
    { "step": "Step 4: Other conditions ... excluded ...", "hit": false },
    { "step": "Step 5: ... orbital cellulitis and sinusitis ...", "hit": true },
    { "step": "Step 6: Imaging findings matched ...", "hit": false },
    { "step": "Step 7: No evidence ... thrombotic or meningeal ...", "hit": false
},
    { "step": "Final Diagnosis: ... orbital cellulitis secondary to acute
sinusitis ...", "hit": true }
  ],
  "efficiency": 1.0,
  "factulity": 1.0,
  "recall": 0.375
}
```

`recall = 3/8 = 0.375`（8 条 GT 步中 3 条 hit）。

Step 6 · gemini2-ft 评估落盘（957 batch 真跑）

同一病例 PMC11625232，换模型 `gemini2-ft`；GT 不变，合并时取：

```
case_data["results"] = oracle["PMC11625232"]["gemini2-ft"]
```

Judge 比较 `generate_case.diagnosis_results` 与 `results.content` 的 `### Answer:` 段。

	文本
GT	Orbital cellulitis secondary to acute sinusitis with frontal epidural empyema.
Pred	Orbital Cellulitis with Frontal Epidural Empyema secondary to Maxillary Sinusitis

`reasoning_results_qwen_judge_paper_957/gemini2-ft/PMC11625232.json`（截断；Accuracy 另存 `acc_results_.../`，仅多 `accuracy` 字段）：

```
{
  "raw_case": "Neurological and orbital complication of acute sinusitis...（原文全
文，评估不读）",
  "generate_case": {
    "case_summary": "- **Patient Information:** 13-year-old male ...",
    "differential_diagnosis": "1. **Orbital Cellulitis:** ...",
    "final_diagnosis": "The patient was diagnosed with **orbital cellulitis
secondary to acute sinusitis** ...",
    "diagnosis_results": "Orbital cellulitis secondary to acute sinusitis with
frontal epidural empyema."
  },
  "body_category": ["Brain and Nerves"],
  "disorder_category": ["Infections"],
  "receive_time": "2024-10-9",
  "checked_rare_disease": [],
```

```
  "id": "PMC11625232",
  "results": {
    "content": "### Resoning:\n<step 1> The patient is a 13-year-old male
presenting with severe left eye pain...\n...<step 7> ... Maxillary
Sinusitis.\n\n### Answer:\nOrbital Cellulitis with Frontal Epidural Empyema
secondary to Maxillary Sinusitis"
  },
  "reasoning_eval": [
    {
      "step": "The patient is a 13-year-old male presenting with severe left eye
pain, eyelid edema, erythema, and localized warmth in the left eye. These are
classic signs of orbital inflammation or infection.",
      "efficiency": "Reasoning",
      "factulity": true,
      "judgment_path": [{ "judgment": "Correct", "keywords_to_search": "None" }]
    },
    {
      "step": "The history of recent severe nasal congestion and headaches
suggests a possible link between the eye symptoms and a sinus infection...",
      "efficiency": "Reasoning",
      "factulity": true,
      "judgment_path": [{ "judgment": "Correct", "keywords_to_search": "None" }]
    },
    "... 共 7 步，均为 Reasoning + factulity=true ..."
  ],
  "gt_reasoning_eval": [
    { "step": "<Step 1> Orbital Cellulitis was strongly suspected given the
patient's symptoms of severe eye pain, erythema, eyelid swelling...", "hit": true
},
    { "step": "<Step 2> Acute Sinusitis was evident on imaging with inflammatory
obliteration of the maxillary sinus...", "hit": true },
    { "step": "<Step 3> A small epidural empyema was identified by NCCT and
confirmed by NCMRI...", "hit": false },
    { "step": "<Step 4> Other conditions such as cavernous sinus thrombosis or
meningitis were excluded...", "hit": false },
    "... 共 9 步 ..."
  ],
  "efficiency": 1.0,
  "factulity": 1.0,
  "recall": 0.556
}
```


`recall = 5/9 ≈ 0.556`（9 条 GT 步中 5 条 hit）。

Accuracy 落盘（`acc_results_qwen_judge_paper_957/gemini2-ft/PMC11625232.json`）：结构同上 GT + `results`，**无** `reasoning_eval` / `gt_reasoning_eval`，仅追加：


```
{
  "...GT + results 同上...",
  "accuracy": true
}
```

Judge 认为 pred 与 `diagnosis_results` 语义等价 → `accuracy: true`。

数据流小结

```
case_summary ──► 模型推理 ──► results.content (+ deepseek 另有 thinking_process)
                                       │
                   ┌───────────────────┼───────────────────┐
                   ▼                   ▼                   ▼
          deepseek: split 5 步      Answer 段          (thinking 默认不用)
                   │                   │
          Eff/Fact/Comp              Accuracy（本例 Step 6 用 gemini2-ft）
                   │                   │
      reasoning_eval[] +          acc_results/.../gemini2-ft/
      gt_reasoning_eval[] +       accuracy: true
      efficiency/factulity/recall
```

---

8. 已知拼写 typo

MedR-Bench 原始 benchmark / 评估代码中有两处历史拼写错误，读数据或写脚本时需按实际**字段名**处理，不要自行“纠正”。

实际写法	正确拼写	来源	说明
`factulity`	factuality	**评估 code**	落盘 JSON 字段名；`reasoning_eval.py`、`oracle_diagnose_reasoning.py` 等均使用此拼写。论文指标名 **Factuality** 本身无误。
`### Resoning:`	`### Reasoning:`	**推理 prompt 模板**	`src/Inference/instructions/oracle_diagnose.txt` 及 HuggingFace 论文推理输出 `oracle_diagnosis.json` 均沿用此格式。评估 code 在 `split_reasoning` 中**同时兼容** `Reasoning` 与 `Resoning` 两种标记（见 `src/Evaluation/utils.py`）。